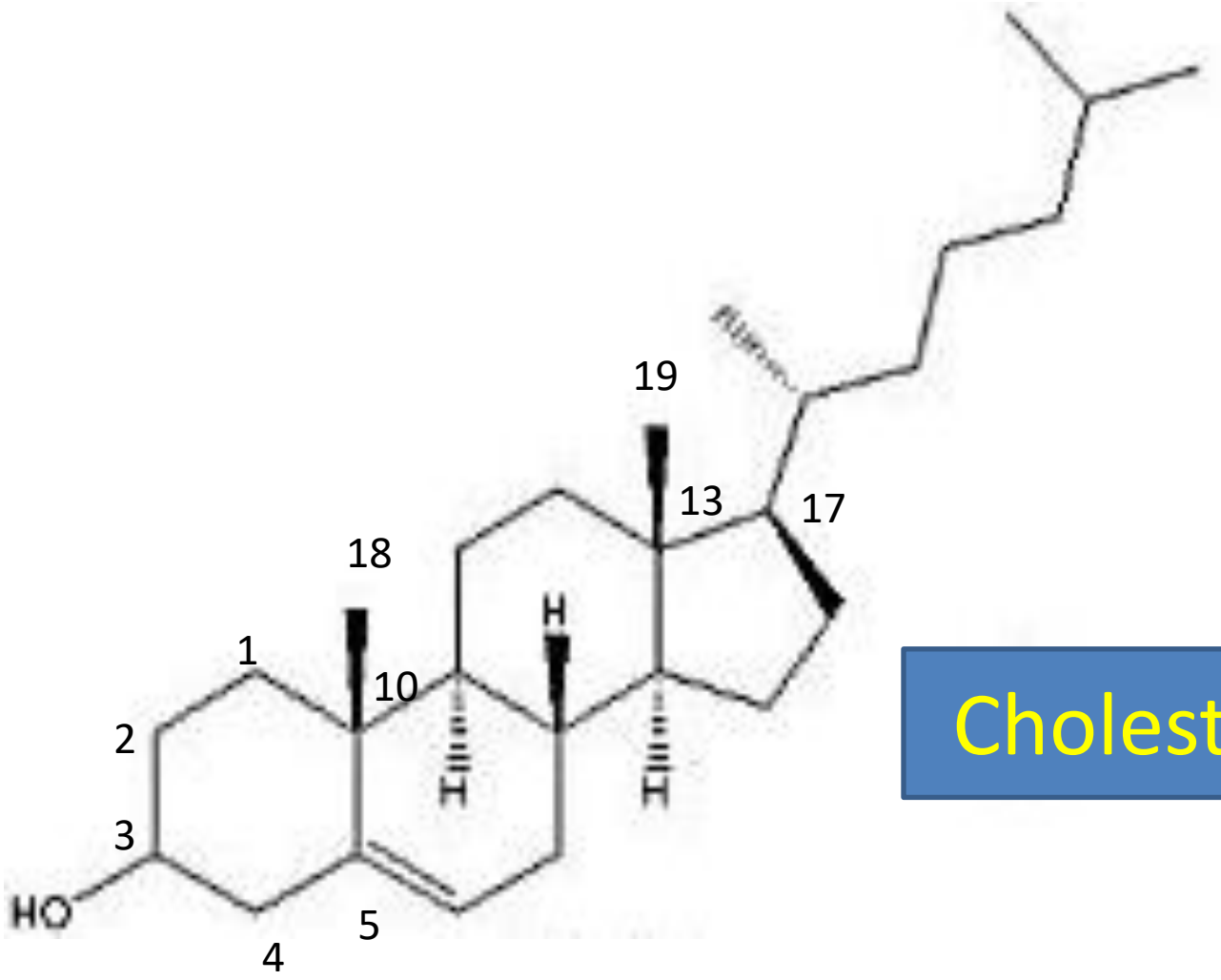
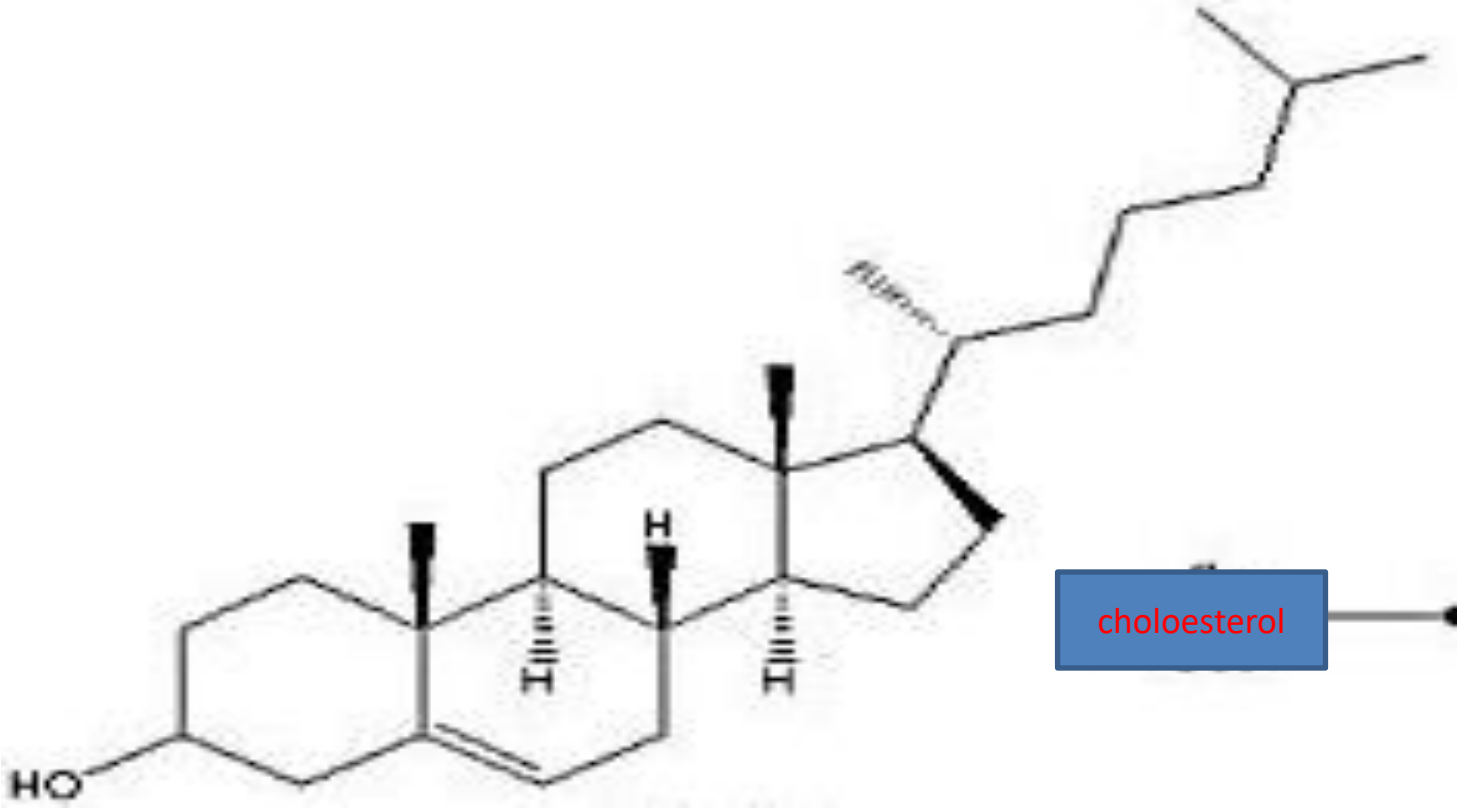
Positions of the hydroxyl group and double bond

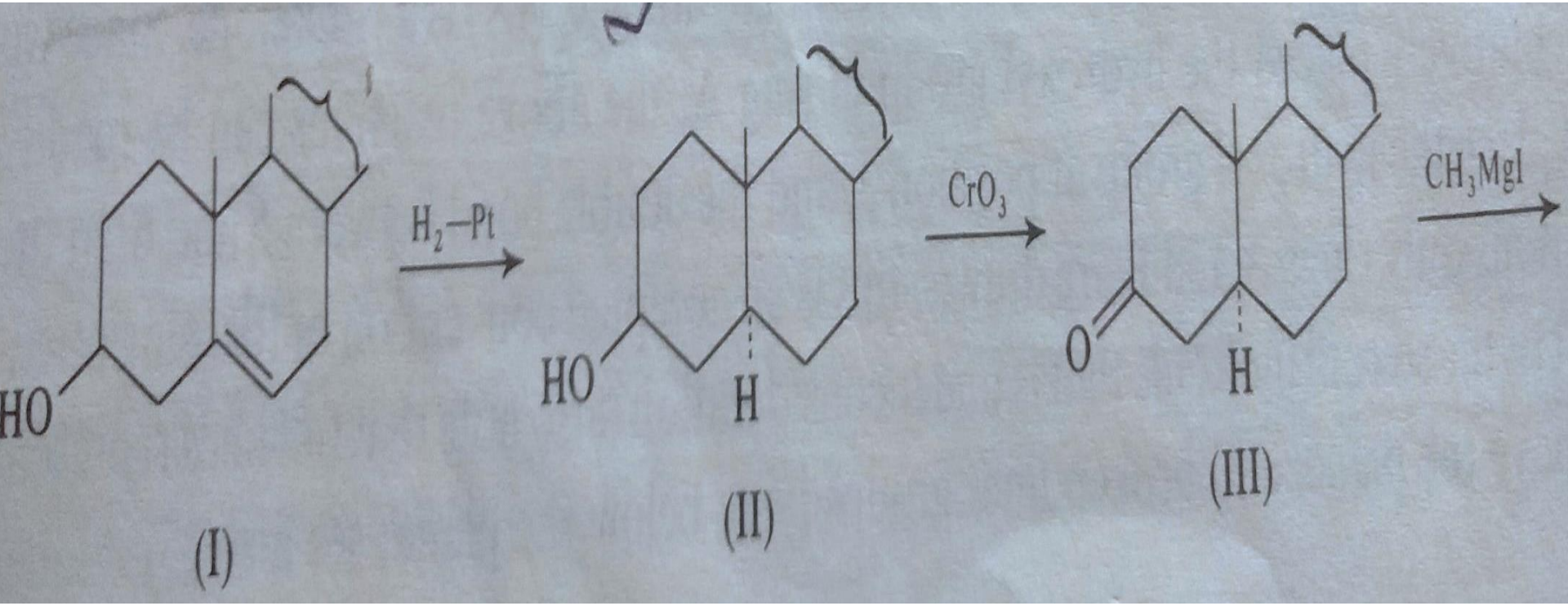
Positions of the angular methyl groups



Cholesterol

- Cholesterol ; $C_{27}H_{46}O$ (I) $\xrightarrow{H_2\text{-}Pt}$
- Cholestanol; $C_{27}H_{48}O$ (II) $\xrightarrow{CrO_3}$
- Cholestanone; $C_{27}H_{46}O$ (III) $\xrightarrow{Zn\text{-}Hg\ (HCl)}$
- Cholestane; $C_{27}H_{48}$ (IV):





Cholesterol Cholestanol Cholestanone

Positions of the hydroxyl group and double bond

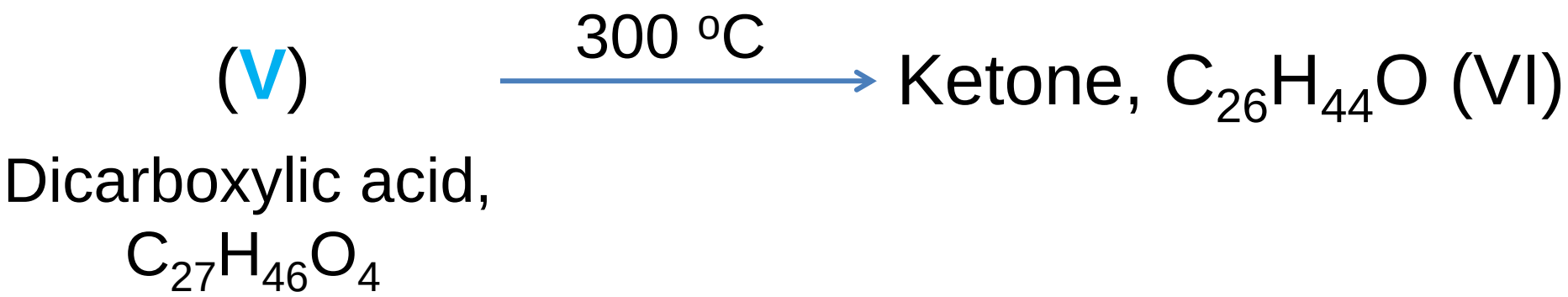
- Cholestanone, $C_{27}H_{46}O$ (III) $\xrightarrow{HNO_3}$ Dicarboxylic acid, $C_{27}H_{46}O_4$ (**V**)

(**V**) has same number of Carbon as the ketone (III) from which it is derived, hence it is in the ring.

Pyrolysis gives ketone with loss of one carbon, hence according to Blanc's rule (**V**) is either a 1,6 or 1,7-dicarboxylic acid.

(**V**) Dicarboxylic acid, $C_{27}H_{46}O_4$ $\xrightarrow{300\ ^oC}$ Ketone, $C_{26}H_{44}O$ (VI)

3 six –membered and 1 five-membered ring possesses; hence dicarboxylic acid (**V**) must be obtained by opening of ring A, B, or C and consequently the –OH is in ring A, B, or C.



Positions of the hydroxyl group and double bond

- Cholesterol ; $C_{27}H_{46}O$ **(I)** $\xrightarrow{H_2O_2,\ /\ CH_3CO_2H}$

Double bond is hydroxylated

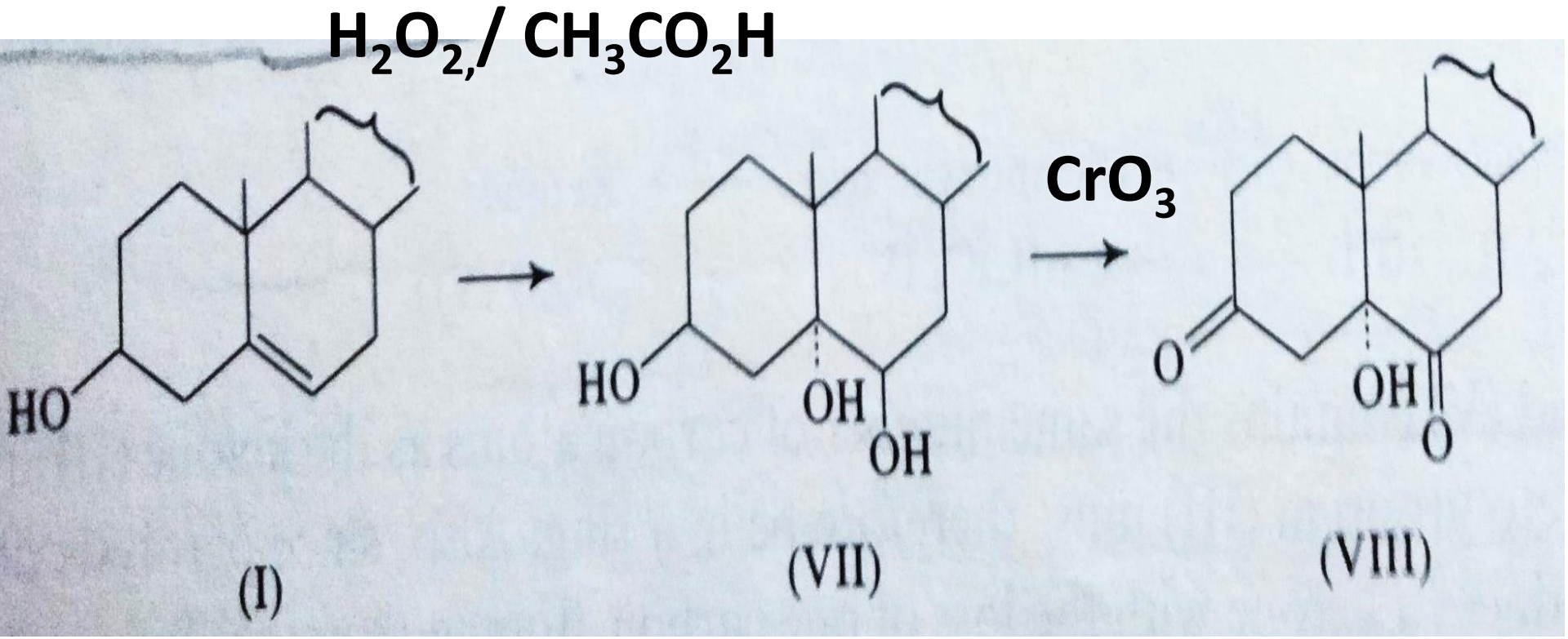
Cholestanetriol;$C_{27}H_{48}O_3$

(**VII**) $\downarrow$ CrO_3

Hydroxycholestanedione

$C_{27}H_{44}O_3$(**VIII**)

Two hydroxyl groups are oxidized hence 2°



❖Formation of **VII** indicates that double bond is hydroxylated

❖Among three –OH, only two is oxidized to produce (**VIII**), so they are 2° alcoholic group; one from cholesterol, 3rd one resistant to oxidation, might be 3° alcohol.

Hydroxycholestanedione $\xrightarrow[(ii)Zn\text{-}CH_3CO_2H]{(i)\ -H_2O}$

$C_{27}H_{44}O_3$(**VIII**)

Cholestanedione

$C_{27}H_{44}O_2$(**IX**)

Dehydration of (**VIII**) by heating in vacuo and subsequent reduction of the double bond forms (**IX**).

$\downarrow$ **CrO_3**

Tetracarboxylic acid,

$C_{27}H_{44}O_8$(X)

Without loss of carbon atoms;
two ketone groups in X must be in different ring;

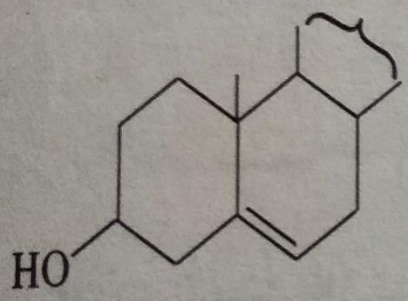
If they were in the same ring C would lost, X not obtained

❖Dehydration of (**VIII**) by heating in vacuo and subsequent reduction of the double bond forms (**IX**).

❖Upon oxidation (**IX**) gives a tetracarboxylic acid (**X**) without loss of carbon atoms. Thus the two keto groups in (**IX**) must be in different rings.

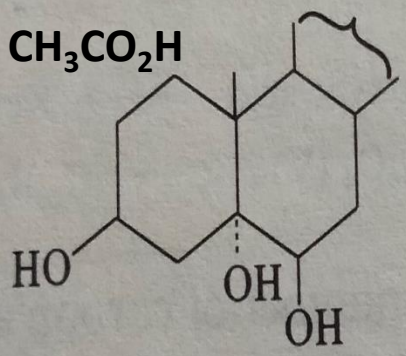
had they been in the same ring, then carbon would have been lost and (**X**) not obtained.

So, **-OH and '=' bond must be in different rings**.



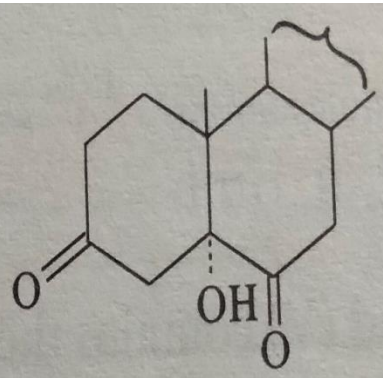
(I)

H_2O_2,/ CH_3CO_2H



(VII)

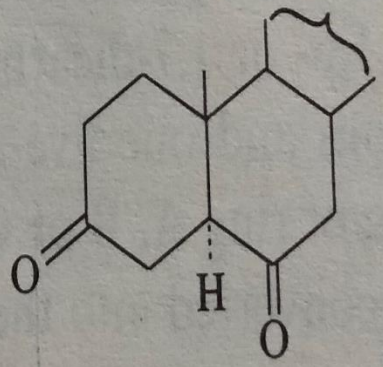
CrO_3



(VIII)

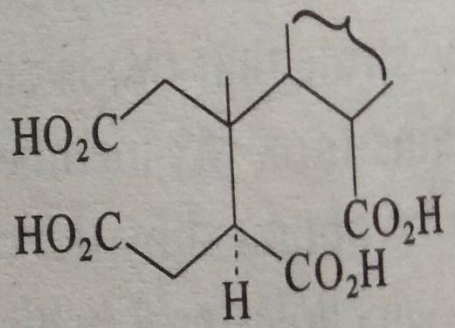
(i) $-H_2O$

(ii) $Zn\text{-}CH_3CO_2H$



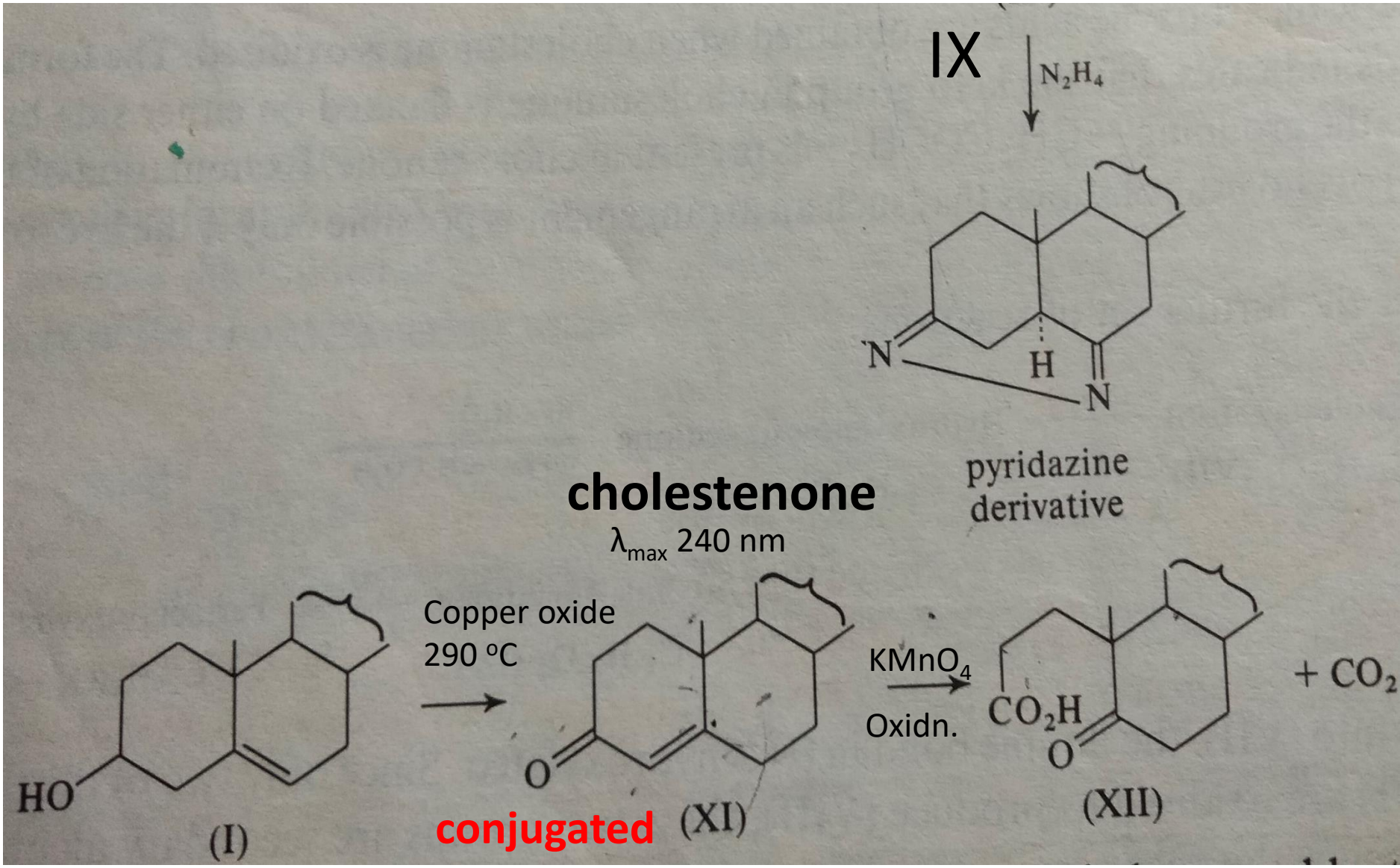
(IX)

CrO_3

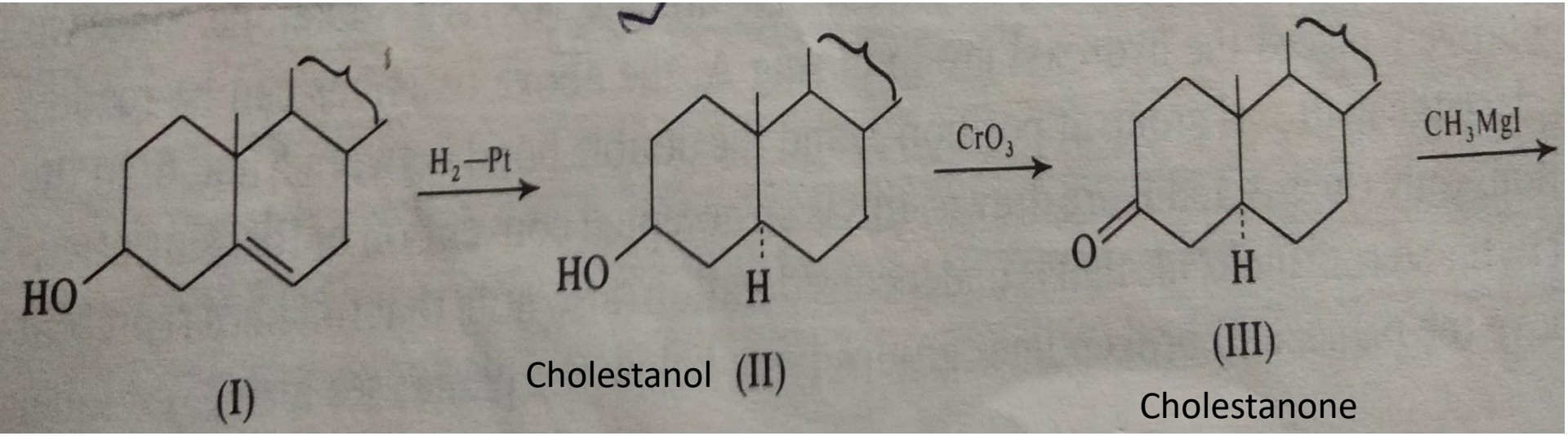


(X)

N_2H_4



The double bond in cholesterol migrates in the formation of XI;
Formation of **XII** indicates that the keto and double bond of XI in the same ring

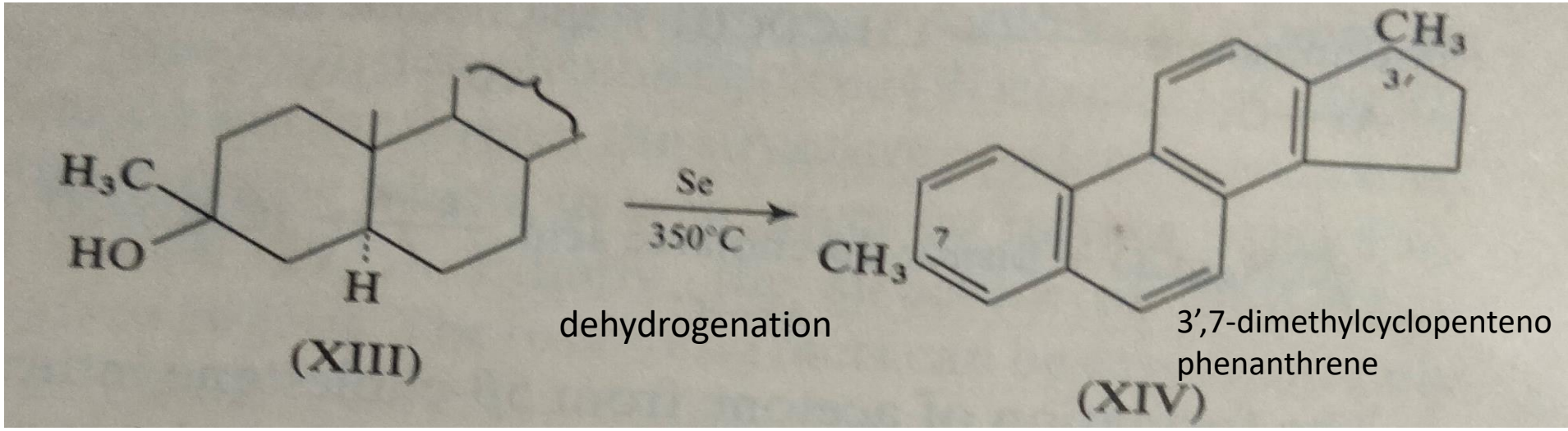


(I)

Cholestanol (II)

(III)

Cholestanone



(XIII)

dehydrogenation

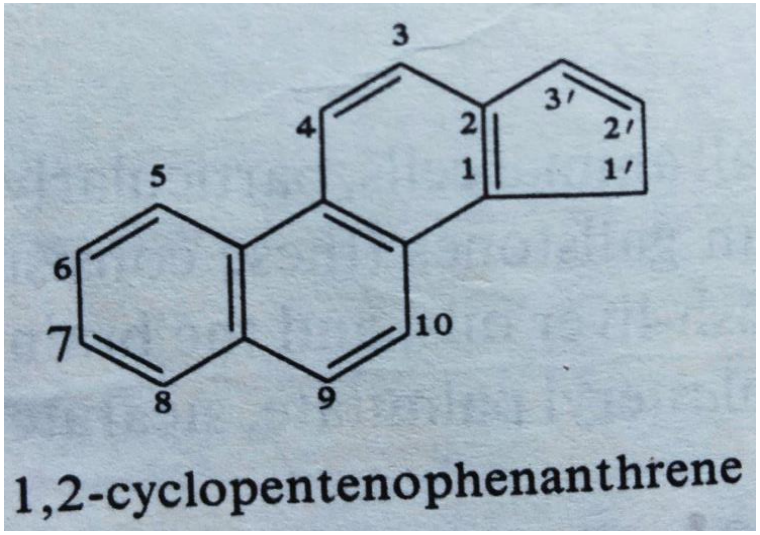
(XIV)

3',7-dimethylcyclopentenophenanthrene

Positions of the angular methyl groups

- Cyclopentenophenanthrene nucleus: 17 C
- Side chain 8 C: 25 C:
- $C_{27}H_{46}O$: rest 2 where?

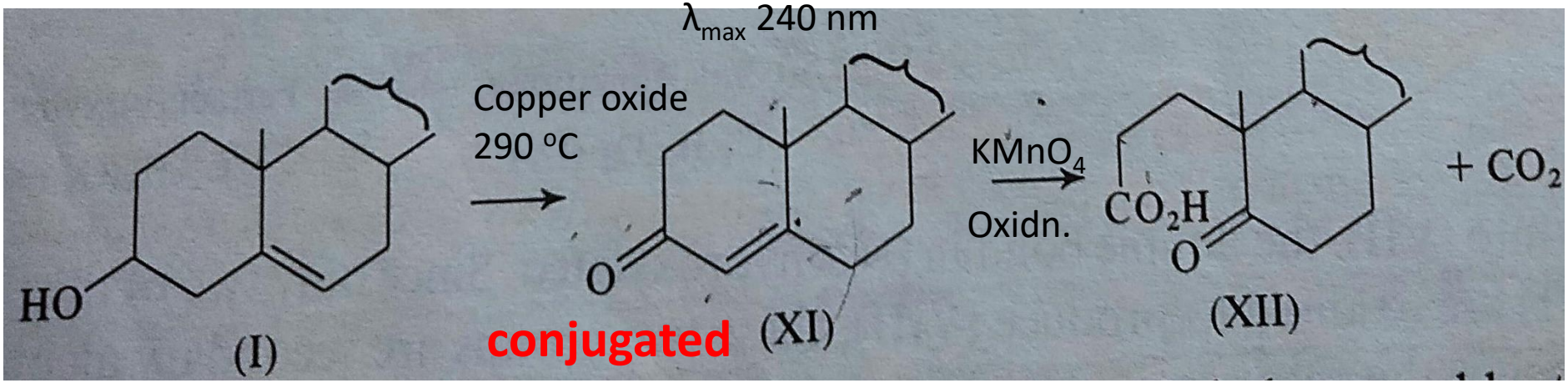
Angular methyl groups

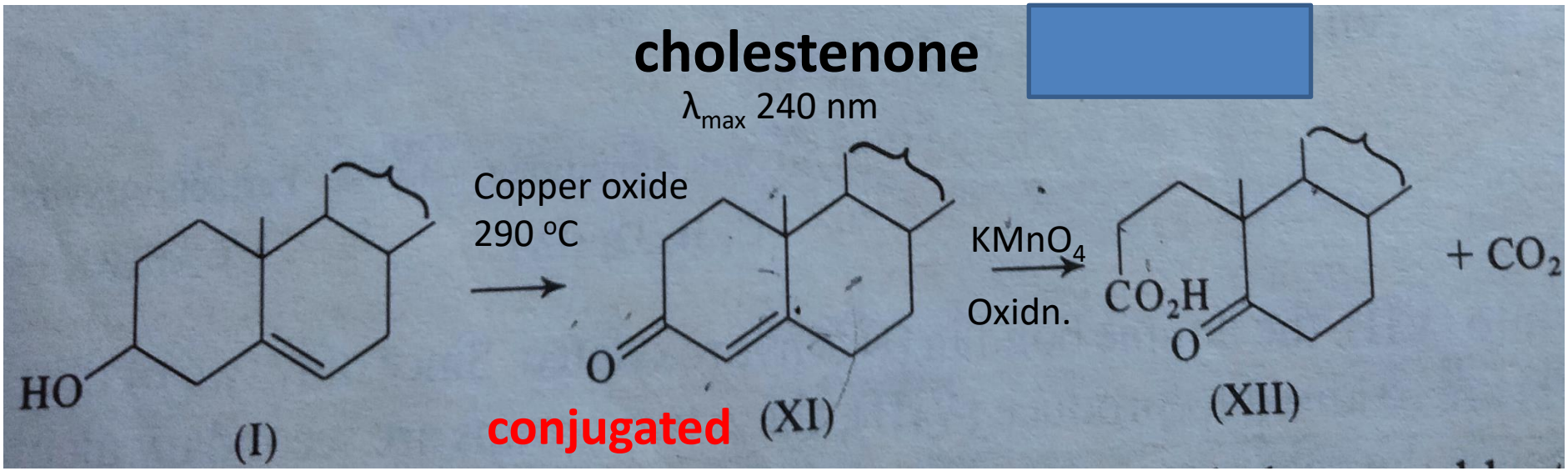


1,2-cyclopentenophenanthrene

cholestenone

λ_{max} 240 nm





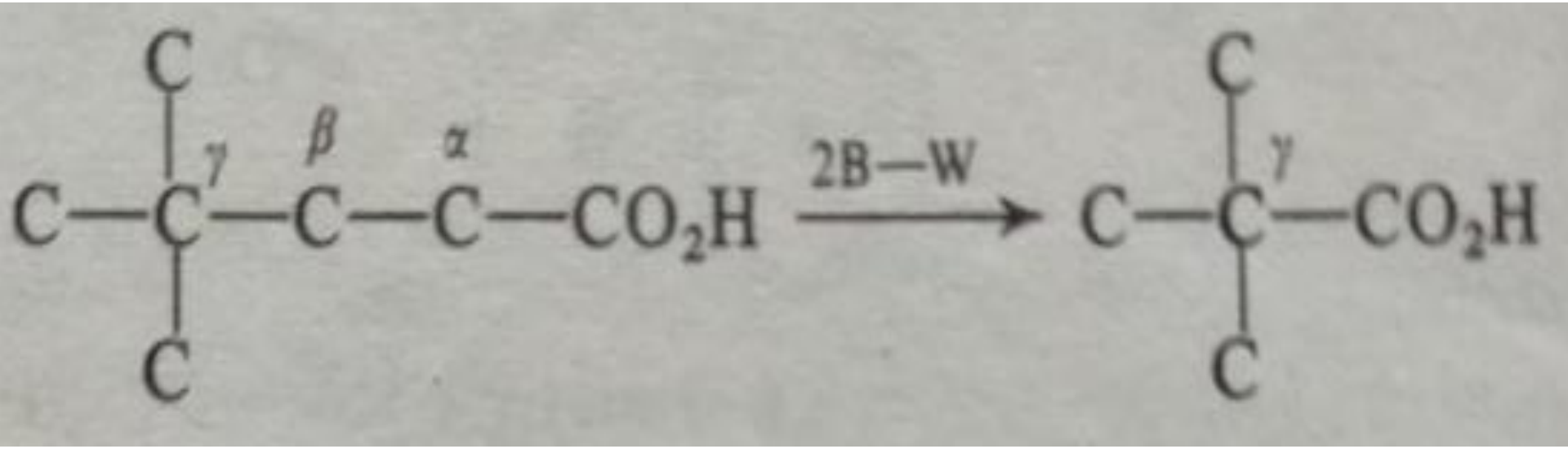
The double bond in cholesterol migrates in the formation of XI; Formation of **XII** indicates that the keto and double bond of XI in the same ring

XII → Clemensen reduction → ? → 2 Barbier- Wieland → Acid

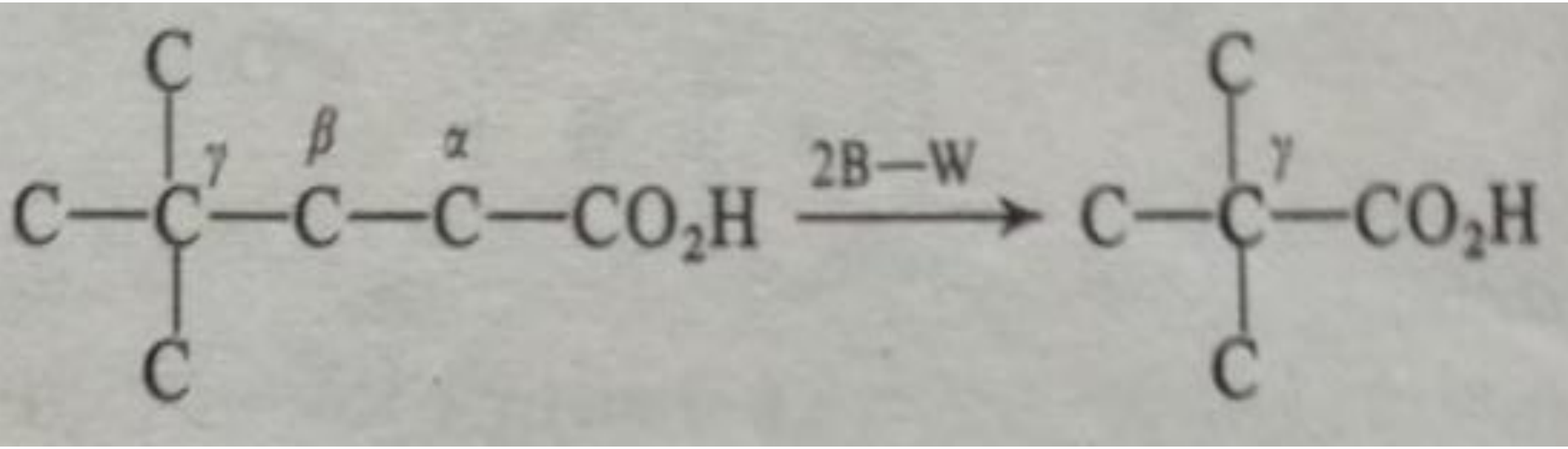
Difficult to esterify
Evolves CO when warmed with sulphuric acid

- Upon Clemensen reduction of XII and followed by two Barbier-Wieland degradation, produces an acid which is very difficult to esterify and evolves CO when warmed with conc. H_2SO_4.

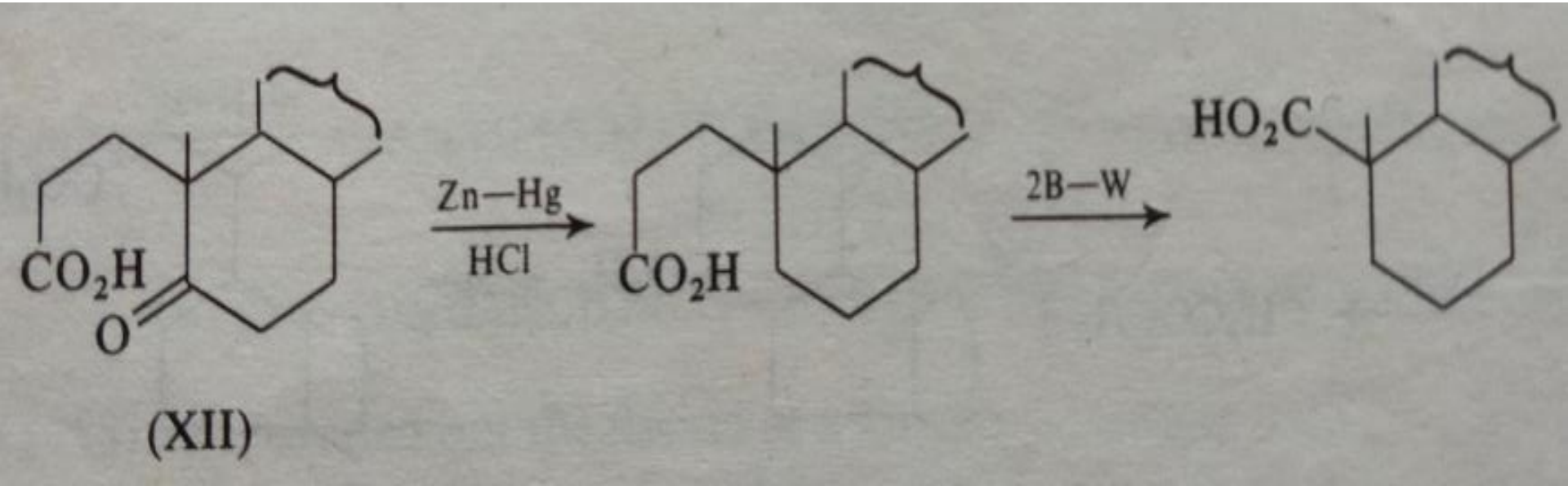
Barbier- Wieland: It's a procedure for shortening the carbon chain of a carboxylic acid by one carbon. Works when the carbon adjacent to the carboxyl is a simple methylene bridge.



Characteristics reaction of an acid containing a carboxyl group attached to a 3° C atom and the side chain in **XII** must be of the type

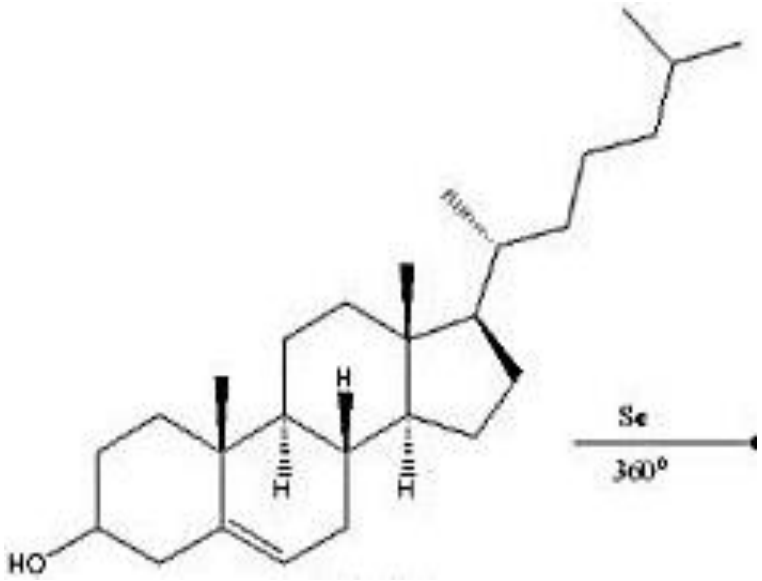


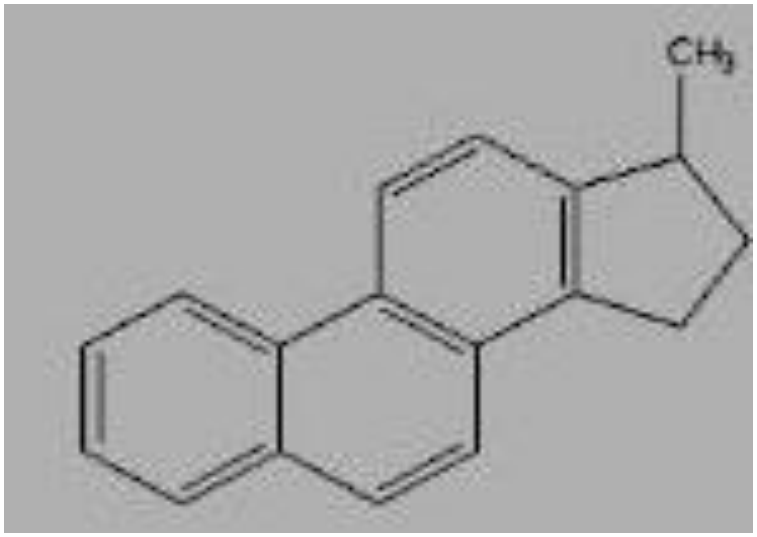
- So, there must be an alkyl group at position 10 in **XII**.
- Either ethyl, -CH_2CH_3 or -CH_3.
- Later findings shows that there is also 1 CH_3 at position 13; Consequently we can say that there is 1 CH_3 at position 10. On this basis the degradation of **XII** is as following:



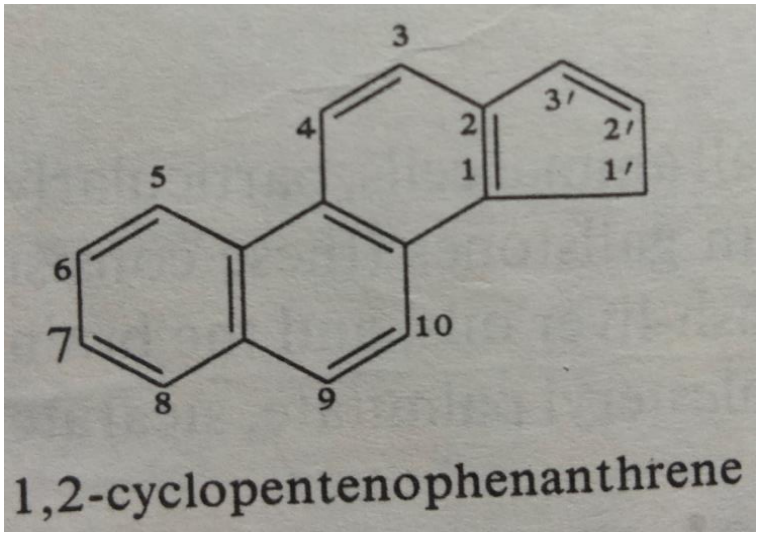
(XII)

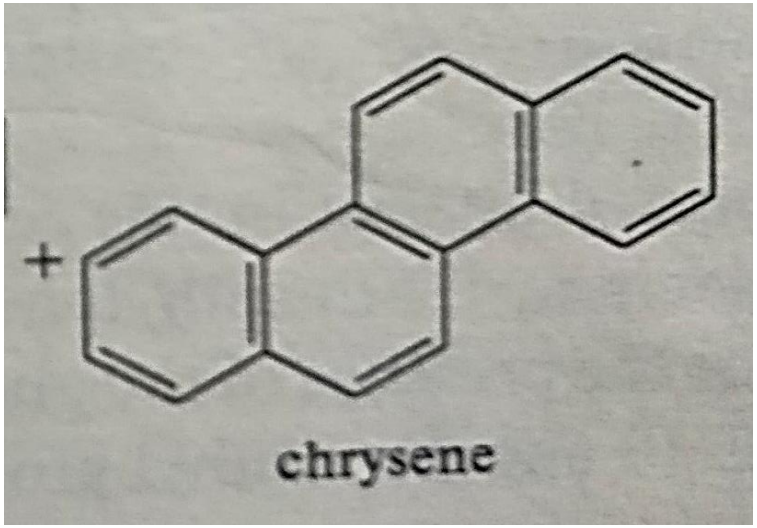
Position of the other angular methyl group is indicated by the following evidence





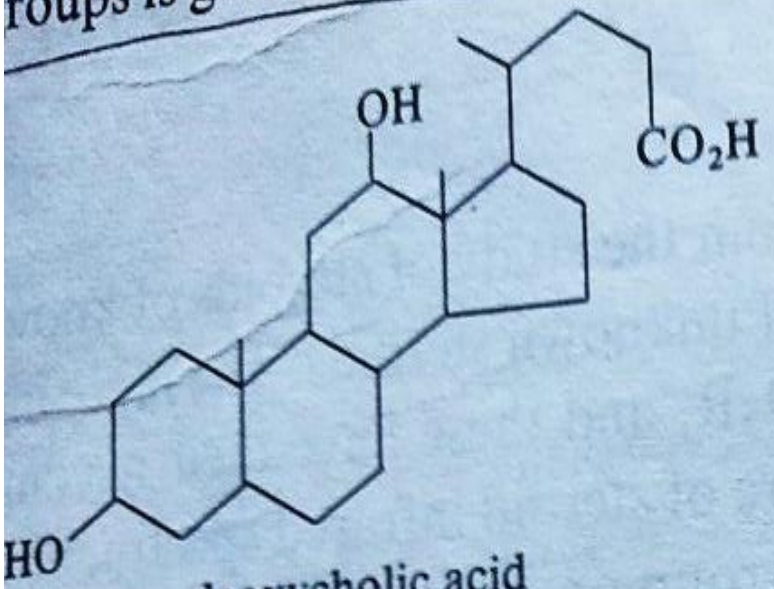
Diels' hydrocarbon





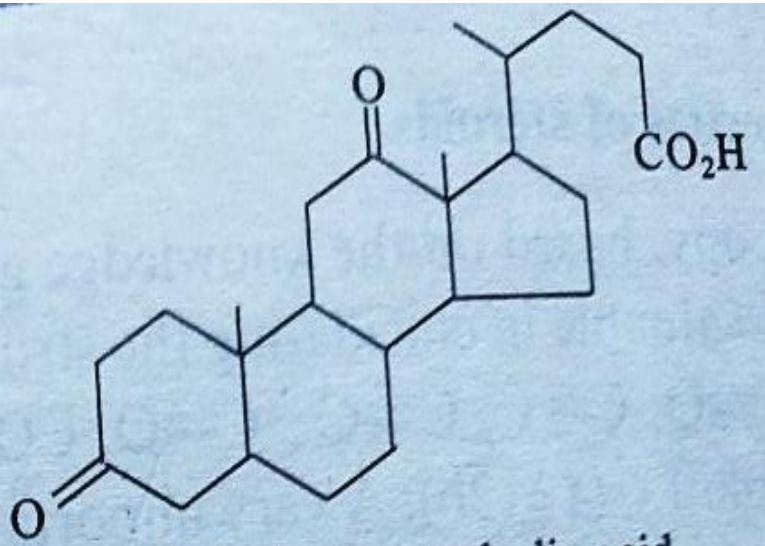
- If Diels' hydrocarbon is the ring skeleton how then chrysene is produced?
- Explanation: there might be a angular methyl group at position 13 which enters into the five membered D ring to form a six-membered ring during dehydrogenation with selenium.
- However this is not so conclusive since……..
- Ring expansion can also occur if the C-14 bears angular methyl group……
- Further support for the two angular methyl groups is given by the following degradative experimental evidence:

roups is given by the



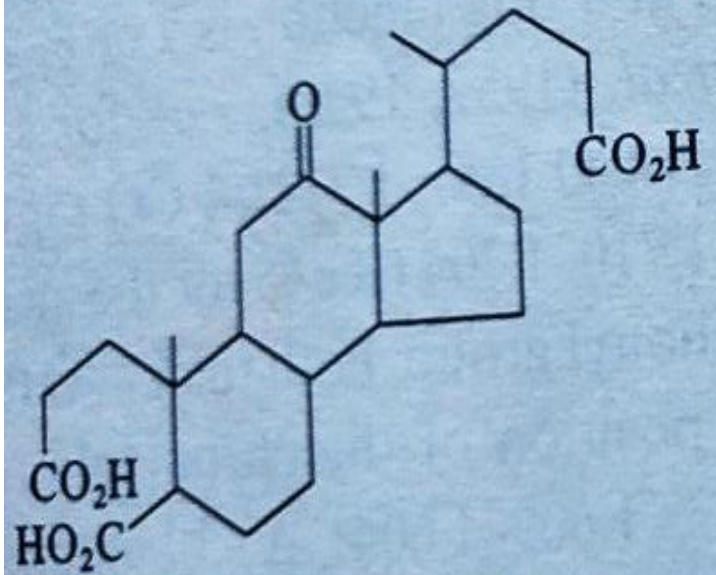
deoxycholic acid

CrO_3



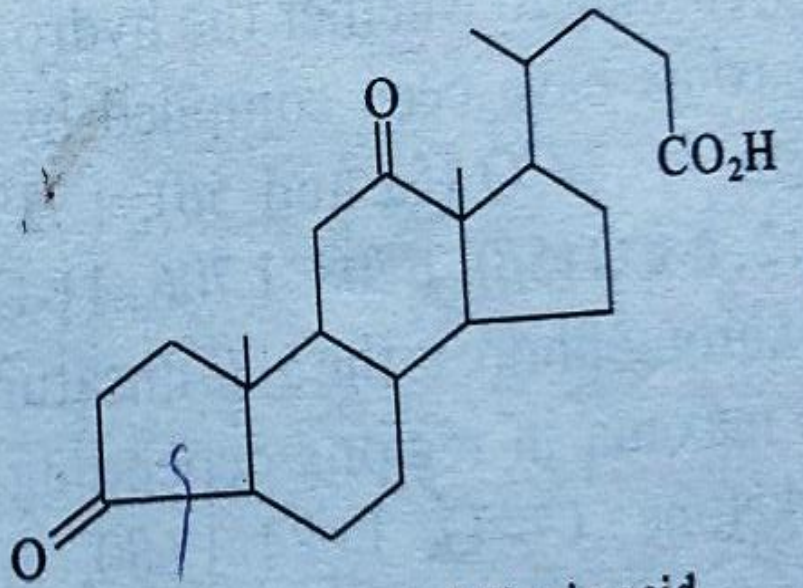
dehydrodeoxycholic acid

HNO_3



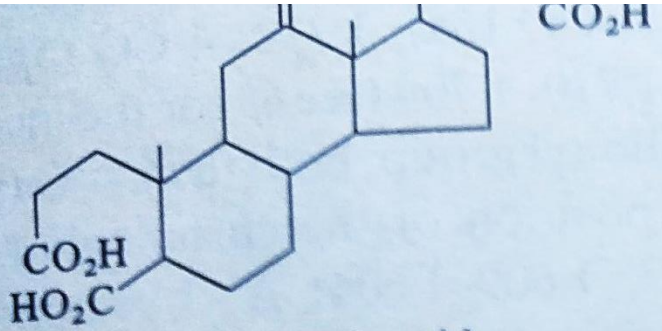
deoxybilianic acid

heat
in vacuo



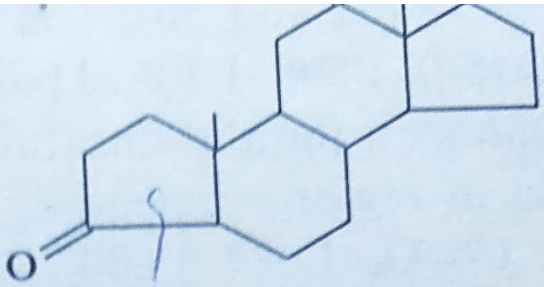
pyrodeoxybilianic acid

$KMnO_4$



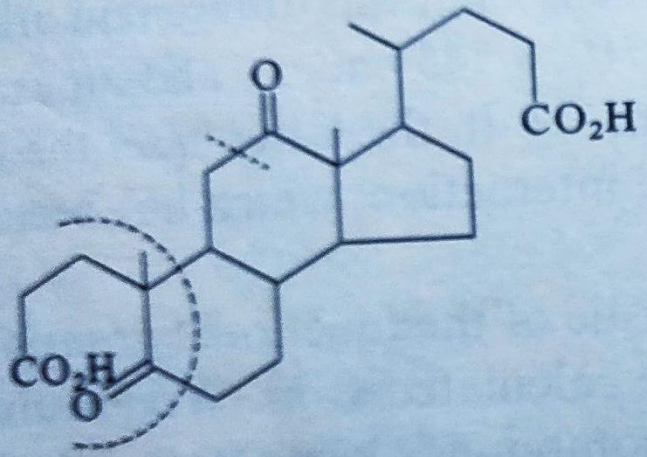
deoxybilianic acid

heat
in vacuo



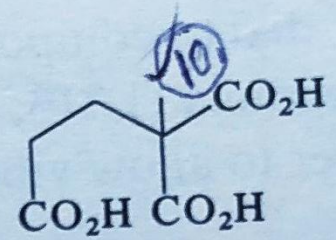
pyrodeoxybilianic acid

$KMnO_4$



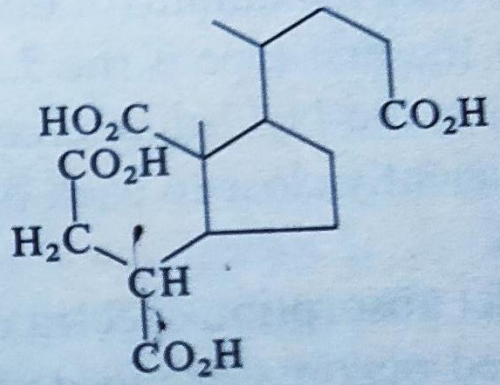
diketo-dicarboxylic acid

HNO_3



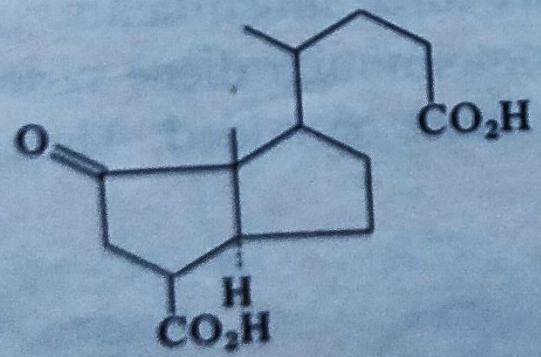
(XVII)

+



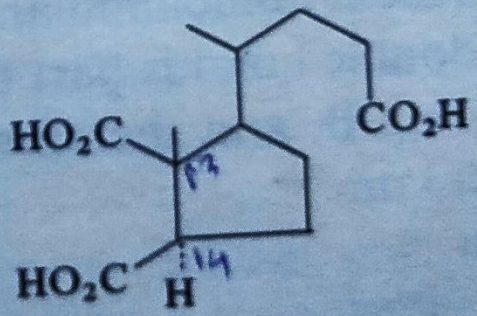
(XVIII)

heat



(XIX)

HNO_3

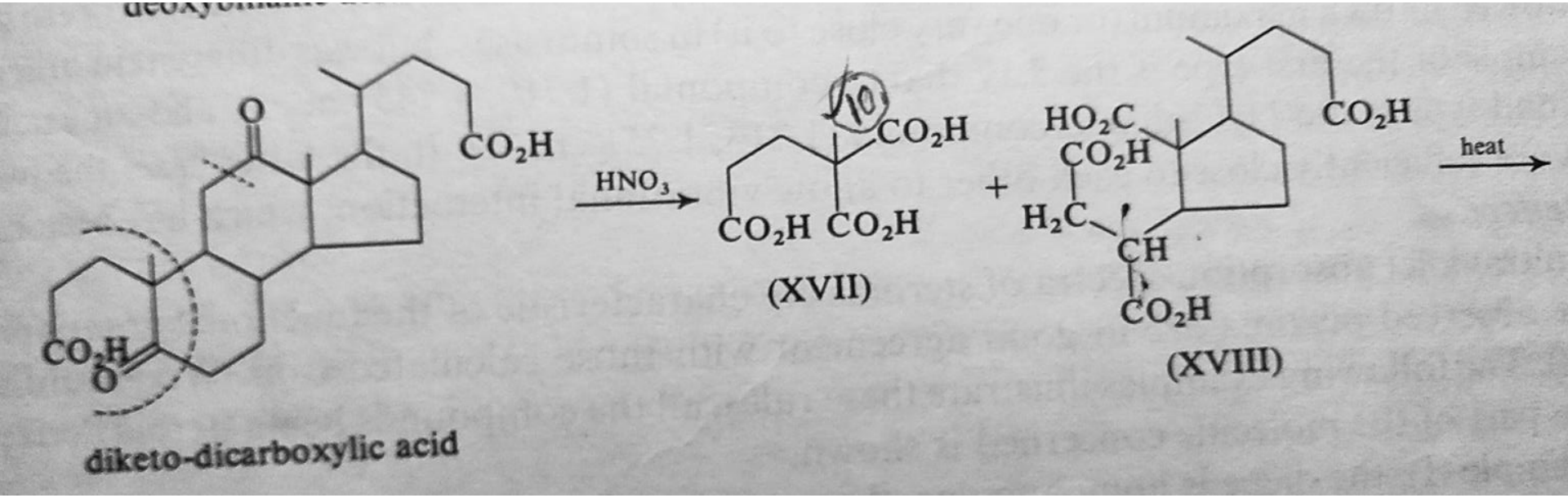


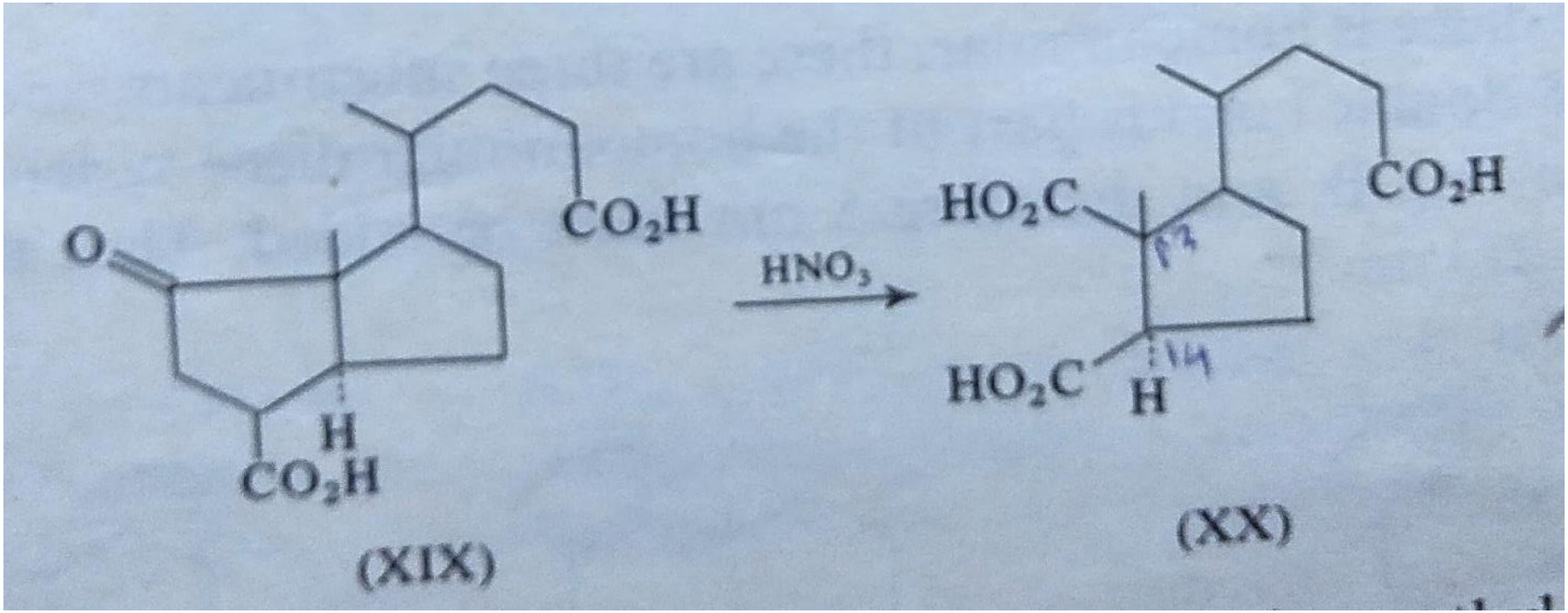
(XX)

XVII was shown to be butane-2,2,4-tricarboxylic acid; thus there is a methyl group at position 10.

XVIII was shown to be a tetracarboxylic acid containing a cyclopentane ring with a side chain:

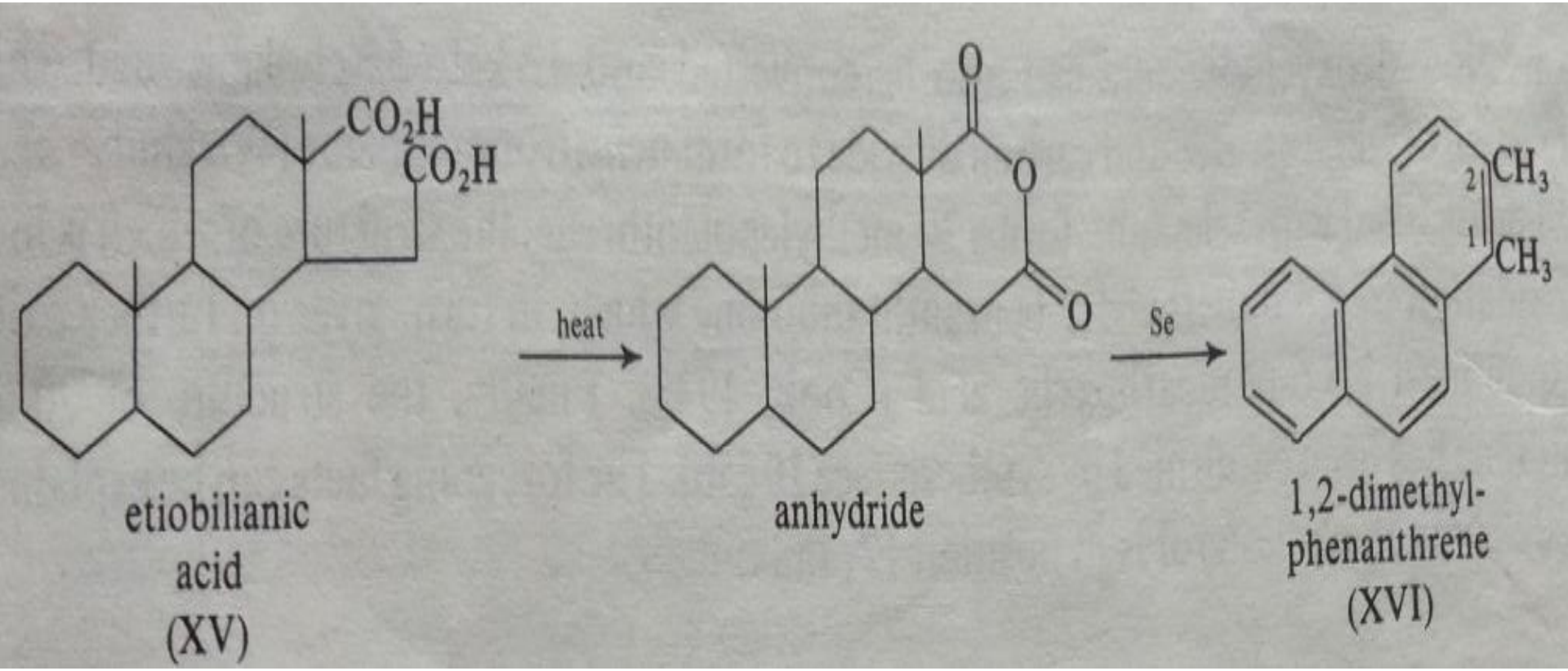
$$-CH(CH_3)CH_2CH_2CO_2H$$





- Thus this compound is derived from ring D.
- XX was also shown to be a tricarboxylic acid containing a cyclopentane ring.
- Furthermore, one carboxyl group in XX was shown to be attached to a 3° carbon atom.
- So, there is a methyl gr. At C-13 or 14.

- XX was shown as trans-configuration, i.e. two carboxyl group s are trans.
- So, its precursor (XIX) must have its **two rings** in the trans configuration (the methyl and H-atom at the ring junction are thus trans.
- Strain involved in cis and trans of (XIX) suggest that cis-(XIX) would have been formed if there were a methyl group at C-14.
- So, the 2nd methyl group at position C-13. further support is following:



- If the methyl gr. were present at C-14 then the above reaction would give 1-methylphenanthrene.